BROOKS PHOTONICS / PUBLIC-DATA CASE STUDY 001

Read-Level Diagnosis of Time-Variable Contamination in a HgCdTe Focal Plane Array

Independent analysis of Hubble WFC3/IR MULTIACCUM measurements

> **The affected ramp contains a strong early spatial gradient followed by a lower-amplitude common-mode tail. The spatial artifact ends before the temporal contamination, and read selection materially changes the reconstructed count-rate product.**

0-3 Spatially anomalous intervals	**0-6** Common-mode intervals	**700 s** Transient duration	**16.1%** Excess / nominal charge

Metric	Result
Data products	2 public IMA ramps and 2 archived FLT products
Detector read structure	16 reads; 14 positive-time intervals
Peak control-relative excess	0.4958 e-/s/pixel
Integrated detector-wide excess	197.26 e-/pixel
Robust tile-median estimate	150.03 e-/pixel
Bootstrap 95% interval	144.61-159.50 e-/pixel

Engineering conclusion. The combined temporal, spatial, control-relative, source-free, bootstrap, and refitting evidence is most consistent with a time-variable external background. It does not support an intrinsic HgCdTe material defect.

Scope. Public-data technical demonstration. No client data are represented. The analysis does not infer composition, carrier lifetime, dark-current mechanism, responsivity, detectivity, or intrinsic material quality.

Prepared by Brooks Photonics | brooks-photonics.com | July 2026

SECTION 01

Data, read-level method, and spatial evolution

The case uses two calibrated WFC3/IR IMA ramps and their archived FLT products from HST program 14037, visit BB. The affected exposure is independently compared with a nominal exposure from the same visit. Exact MAST URIs, byte counts, and SHA-256 hashes are retained in the reproducibility inventory.

For calibrated read rate $R_i(x,y)$ at sample time t_i, cumulative charge is reconstructed as $Q_i = R_i t_i$. Consecutive-read interval rate is then $[Q_i - Q_{i-1}] / [t_i - t_{i-1}]$.

Metric	Result
Chronology	IMA read extensions are sorted by increasing sample time.
DQ policy	All standard 16-bit WFC3 flags are rejected except DATAREJECT=8192.
Why retain DATAREJECT	It is generated by ramp fitting and is diagnostic evidence, not a permanent detector defect.
Spatial statistics	Sigma-clipped full-frame, left-half, and right-half medians.
Control subtraction	Matched interval rate minus nominal control and late-time static offset.

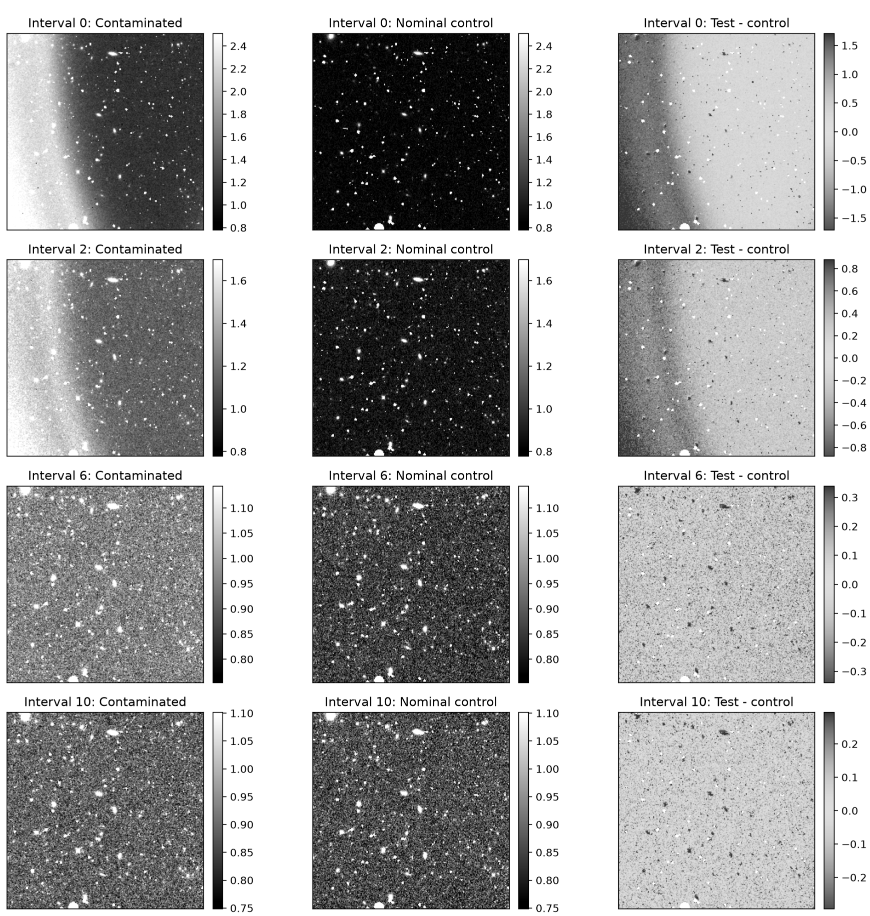

Figure 1. Representative instantaneous interval-rate maps for the affected exposure, nominal control, and matched difference. The strong early gradient transitions into a weaker detector-wide tail.

SECTION 02

Control-relative transient diagnosis

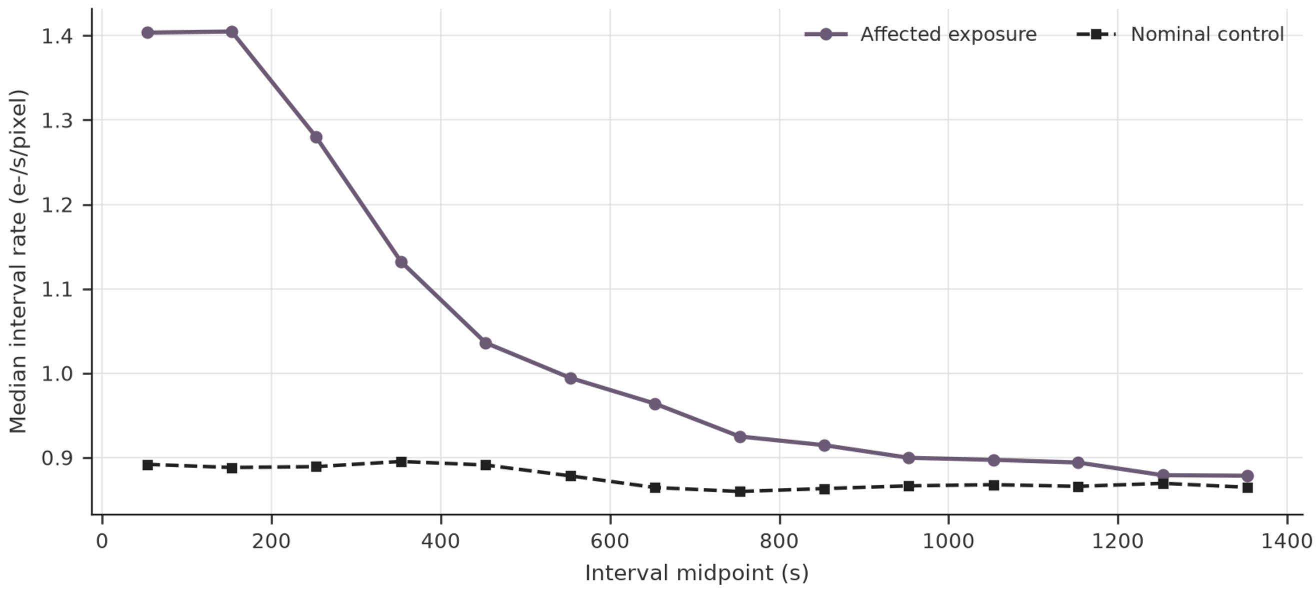

Figure 2. Full-frame interval rate. The affected exposure begins well above the nominal control and decays toward it through the ramp.

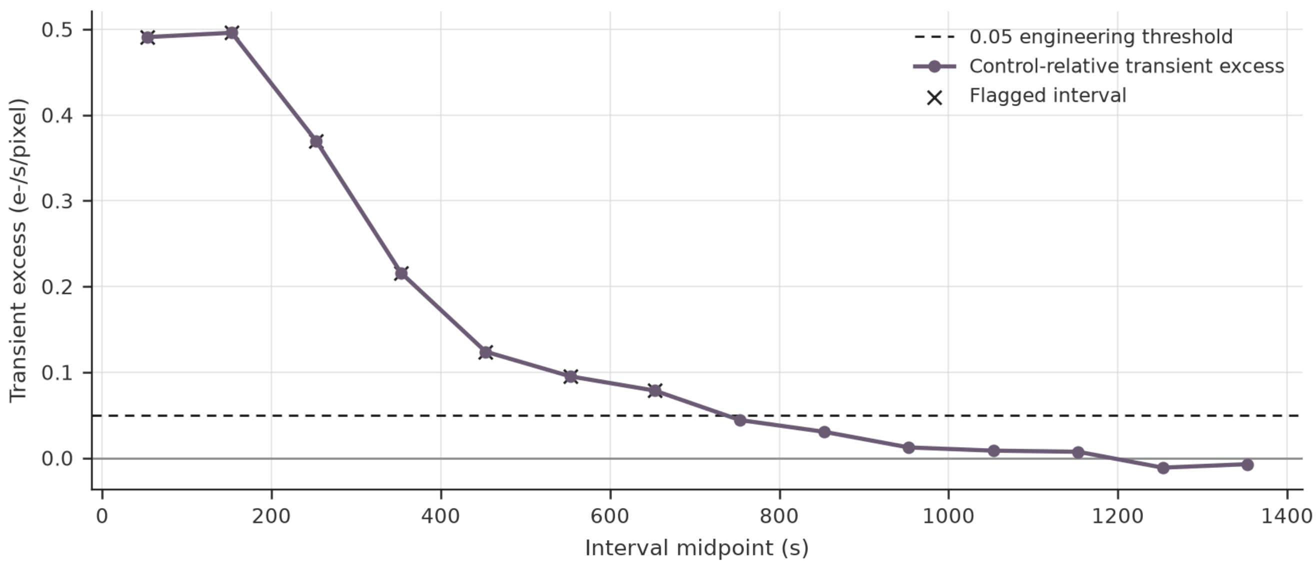

Figure 3. Late-offset-corrected transient excess. Intervals 0-6 exceed the nominal 0.05 e-/s/pixel engineering threshold.

Metric	Result
Spatially anomalous intervals	0-3
Common-mode anomalous intervals	0-6
Peak affected / control asymmetry	53.85% / 1.83%
Late-time static offset removed	0.02096 e-/s/pixel
Pixels ever carrying DATAREJECT	35.80%

SECTION 03

Robustness, source masking, and uncertainty

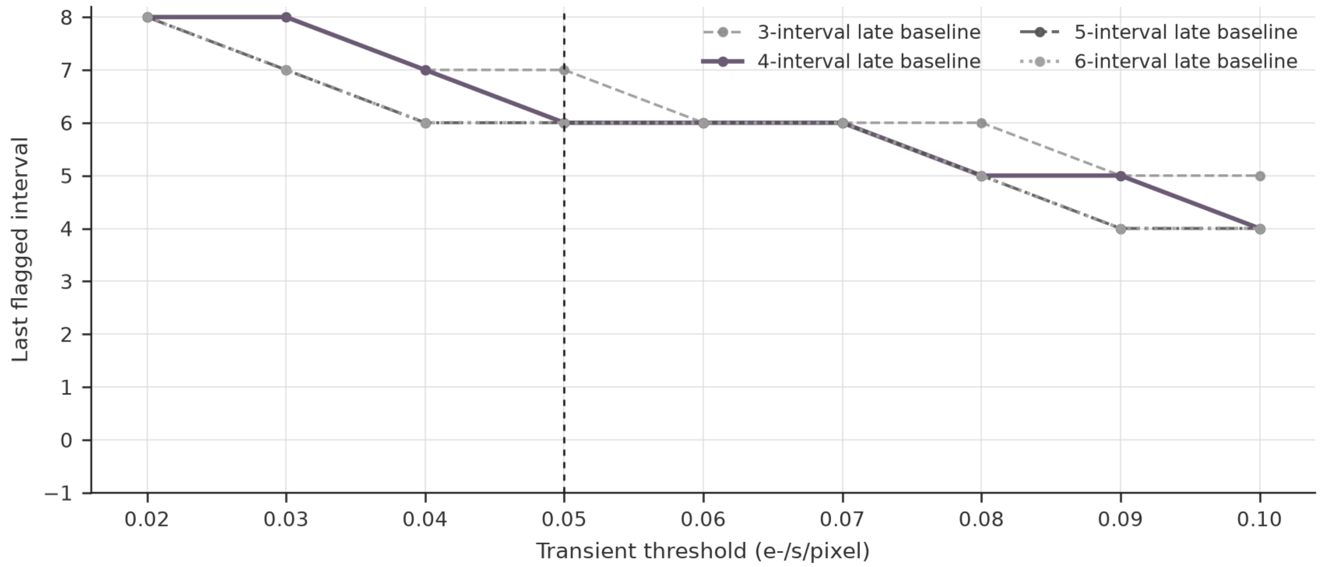

Figure 4. Threshold and baseline sensitivity. Across the complete sweep, the last flagged interval ranges from 4 to 8; the nominal setting classifies interval 6 as the endpoint.

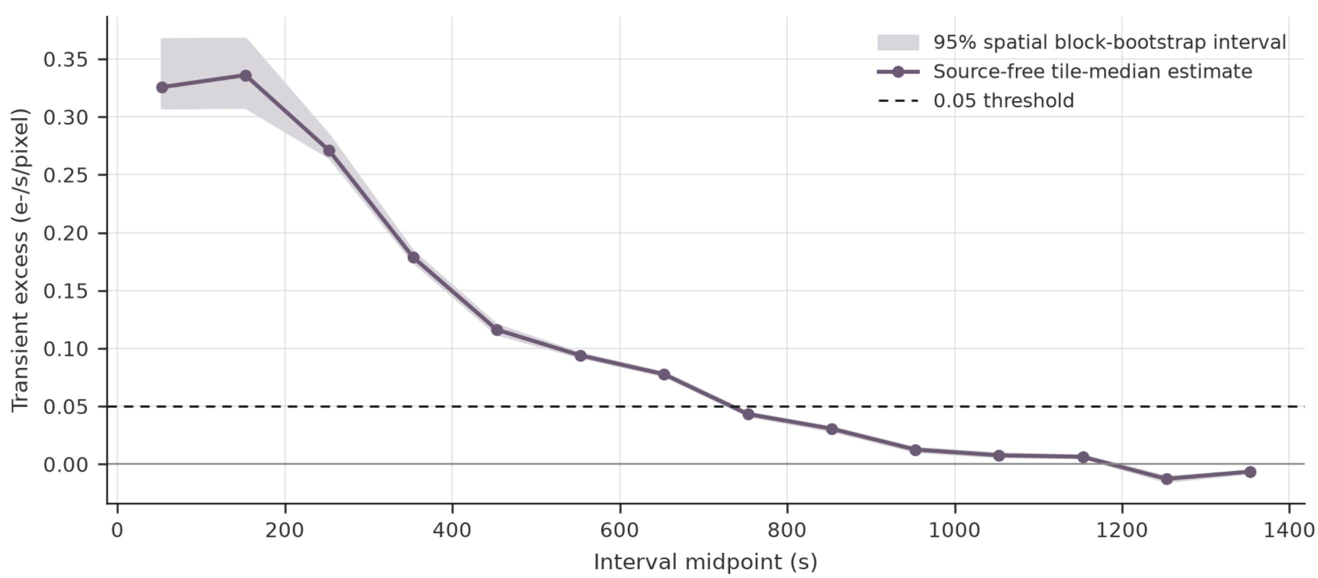

Figure 5. Spatial block-bootstrap interval. All 2,000 resamples preserve the 700.002 s nominal classification duration.

Metric	Result
Source-free pixels retained	94.25%
Source-free classification agreement	100.0%
Spatial tiles / bootstrap replicates	779 / 2000
Integrated excess, 95% interval	144.61-159.50 e-/pixel
Peak excess, 95% interval	0.3089-0.3680 e-/s/pixel

SECTION 04

Independent ramp reconstruction and FLT comparison

A free-intercept, DQ-aware ordinary-least-squares slope is fit to cumulative charge. The post-transient fit begins after interval 6. Earlier excess charge can change the fitted intercept without forcing the later steady-state slope upward.

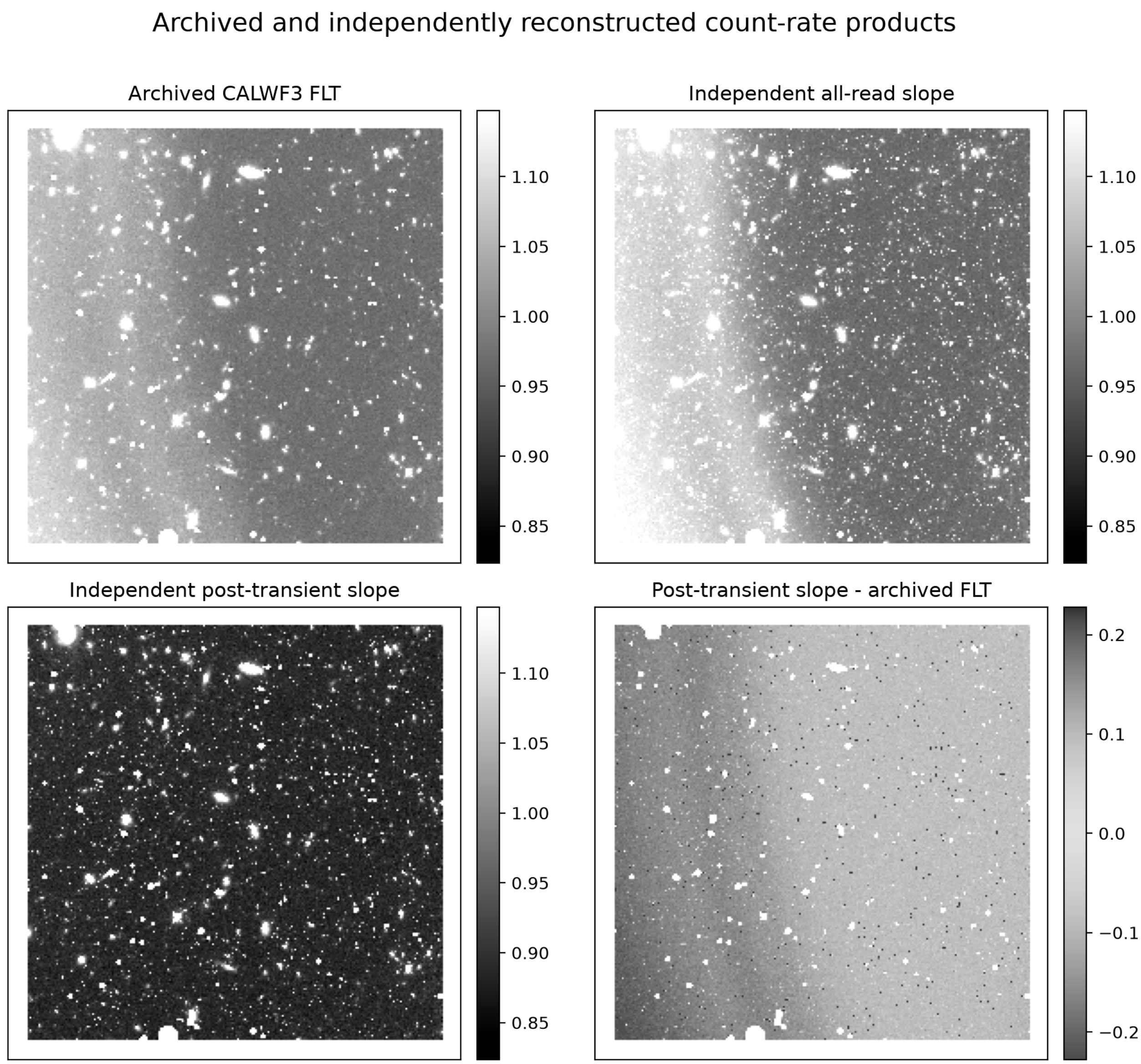

Figure 6. Archived CALWF3 FLT, independent all-read fit, independent post-transient fit, and post-transient-minus-FLT residual. Images are rendered in grayscale for print inspection.

Metric	Result
Post-transient fit begins	t = 702.934 s
Archived FLT source-free median	1.00296 e-/s/pixel
Independent all-read median	0.99992 e-/s/pixel
Independent post-transient median	0.89337 e-/s/pixel
Post-transient minus FLT median residual	-0.11108 e-/s/pixel
Post-transient vs FLT residual NMAD	0.05243 e-/s/pixel

SECTION 05

Engineering interpretation and recommended disposition

> **The obvious spatial gradient ends before the common-mode transient. A spatial-only diagnostic would terminate the anomaly too early and retain intervals with measurable control-relative excess.**

Hypothesis	Assessment
Time-variable external background	Most consistent with the observed temporal decay and spatial evolution.
Stable gain or flat-field nonuniformity	Disfavored; it should persist through the ramp and appear similarly in the control.
Intrinsic dark-current nonuniformity	Strongly disfavored; the detector-wide pattern decays during one exposure.
Readout-quadrant offset	A secondary contribution is possible but does not explain the smooth decay.
Isolated cosmic-ray event	Cannot explain a detector-wide, smoothly decaying background.

Recommended disposition

1. Inspect consecutive-read maps before accepting the final slope product. 2. Classify spatial and common-mode effects separately. 3. Use a matched control or stable late-ramp baseline. 4. Report threshold and baseline sensitivity. 5. Refit after excluding the transient regime and compare the result with the archived product.

Interpretation boundary. Do not infer a detector material defect until measurement-condition explanations have been rejected. This case does not estimate composition, lifetime, responsivity, detectivity, or intrinsic material quality.

Primary sources

STScI WFC3/IR IMA Visualization Tools; STScI scattered-light correction notebook; WFC3 Data Handbook, file structure and IR calibration sections; public products obtained from MAST.

Brooks Photonics | brooks-photonics.com | Independent physics-based infrared detector analysis